AI Security Patch Impact Analyzer

Project: AI-Security-Patch-Impact-Analyzer

Changed Files

File
app.py

Changed Functions

Change Type	File	Function
Added	app.py	security_check
Removed	app.py	paraphrase_text
Modified	app.py	home

Dependency Summary

File	Function	Dependencies
app.py	autocorrect	None
app.py	simplify_text	None
app.py	security_check	None
app.py	home	None

AI Security Analysis

Risk Level

MEDIUM – The patch modifies the existing `home` function, which can change application behavior. While the added `security_check` function and the removal of `paraphrase_text` appear limited in scope, the modification to an existing function warrants a medium■risk classification.

Affected Modules

- `app.py`
- Added: `security_check`
- Removed: `paraphrase_text`
- Modified: `home`

Possible Side Effects

- Changes in `home` may alter the main entry point or routing logic, potentially causing functional regressions or unexpected UI behavior.

- Removing `paraphrase_text` could raise `ImportError` or `AttributeError` in any code that still references it, leading to runtime failures.
- Adding `security_check` does not affect current behavior unless it is invoked elsewhere, but future integration could introduce new security checks that need proper handling.

Testing Suggestions

1. Unit Tests – Extend or update unit tests for `home` to verify that its output and side effects remain correct after modification.

2. Integration Tests – Run end■to■end tests that exercise the routes or features that rely on `home` to ensure overall application flow is unchanged.

3. Dependency Checks – Search the codebase for any references to `paraphrase_text`; add tests to confirm that missing references are handled gracefully (e.g., proper error messages or fallback logic).

4. Security Function Tests – If `security_check` is intended to be called, add unit tests that validate its expected behavior (input validation, return values, etc.).

5. Regression Suite – Execute the full regression test suite to catch any unintended side effects introduced by the modifications.

End of Report